EXERCISE 1: SQL FUNDAMENTALS

EMPLOYEES

Id	First_name	Last_name	department	Salary	Hire_date	City
1	Alice	Green	IT	70000	2020-01-10	Johannesburg
2	Brian	Lee	HR	45000	2019-03-22	Cape Town
3	Cathy	Zulu	Finance	65000	2018-07-18	Durban
4	David	Mokoena	Marketing	50000	2021-11-05	Pretoria
5	Eva	Naidoo	IT	72000	2017-09-30	Johannesburg

(01) Retrieve all columns from the employees table.

→ SELECT *

(02) Find all departments.

→ SELECT DISTINCT department
FROM employees;

department
IT
HR
Finance
Marketing
IT

(03) Retrieve first and last names ordered by salary descending.

⇒ SELECT first_name, last_name
FROM employees
ORDER BY salary DESC;

First_name	Last_name	Salary
Eva	Naidoo	72000
Alice	Green	70000
Cathy	Zulu	65000
David	Mokoena	50000
Brian	Lee	45000

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(04) Retrieve the top 3 Highest-paid employees

⇒
```
SELECT id, first_name, last_name, salary
FROM employees
ORDER BY salary DESC
LIMIT 3;
```

id	first_name	last_name	salary
5	Eva	Naidoo	72000
1	Alice	Green	70000
3	Cathy	Zulu	65000

(05) Find employees in the IT department.

⇒
```
SELECT employee_id, first_name, last_name, department
FROM employees
WHERE department = 'IT';
```

id	first_name	last_name	department
1	Alice	Green	IT
5	Eva	Naidoo	IT

(06) Find employees in Finance with salary > 60000.

⇒
```
SELECT employee_id, first_name, last_name, department, salary
FROM employees
WHERE department = 'Finance' AND salary > 60000;
```

id	first_name	last_name	department	Salary
3	Cathy	Zulu	Finance	65000

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07) Find employees in HR or Marketing

⇒ SELECT employee_id, first_name, last_name, department
FROM employees
WHERE department = 'HR' OR 'Marketing';

id	first_name	last_name	department
2	Brian	Lee	HR
4	David	Mokoena	Marketing

08) Find employees not in IT.

⇒ SELECT employee_id, first_name, last_name, department
FROM employees
WHERE department NOT IN 'IT';

id	first_name	last_name	department
2	Brian	Lee	HR
3	Cathy	Zulu	Finance
4	David	Mokoena	Marketing

09) Find employees in IT, HR or Finance using IN

⇒ SELECT id, first_name, last_name, department
FROM employees
WHERE department IN 'IT', 'HR', 'FINANCE';

id	first_name	last_name	department
1	Alice	Green	IT
5	Eva	Naidoo	IT
2	BRIAN	Lee	HR
3	Cathy	Zulu	Finance

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(10) Find employees in IT with salary > 65 000 and City Johannesburg.

⇒
```
SELECT id, first_name, last_name, department, salary, city
FROM employees
WHERE department = 'IT'
    AND salary > 65000
    AND city = 'Johannesburg';
```

id	first_name	last_name	department	salary	city
1	Alice	Green	IT	70 000	Johannesburg
5	Eva	Naidoo	IT	72 000	Johannesburg

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